Quiz 1.1

Name and Uniqname:

Each question has **one correct answer.** Circle the correct answer only.

Question 1

Which R expression is a Monte Carlo estimate of $\int_0^1 \cos(\pi x/2)\,dx$?

1. `cos(pi * mean(runif(1000)) / 2)`
2. `mean(cos(pi * runif(1000) / 2))`
3. `mean(cos(pi * (1:1000) / 2000))`
4. `mean(runif(1000)) * cos(pi / 2)`

Solution

Answer: 2. A Monte Carlo estimate averages the function evaluated at the draws. Option 1 evaluates the function at the average instead, and $E(g(U))$ is not $g(E(U))$. Option 3 is a grid rule. Option 4 has nothing to do with the integrand.

Question 2

You want to estimate $\int_0^\infty g(x)\,dx$ using draws `x <- rexp(n)` from the Exp(1) distribution. Which estimator is correct?

1. `mean(g(x) * dexp(x))`
2. `mean(g(x))`
3. `mean(g(dexp(x)))`
4. `mean(g(x) / dexp(x))`

Solution

Answer: 4. If X has density f, then $\mathrm{E}\,(g(X)/f(X)) = \int g(x)\,dx$ over the region where f is positive. Dividing by the density is what turns an expectation into the integral we want. Option 2 estimates $\int gf$, which is a different integral.

Question 3

Let `x <- rexp(10000)`. What does `mean(x <= 2)` estimate?

1. $P(X \le 2)$
2. $\mathrm{E}\,(\min(X, 2))$
3. The density of X at 2
4. $\mathrm{E}\,(X) \times P(X \le 2)$

Solution

Answer: 1. `x <= 2` is a vector of indicators for the event $X \le 2$, and the expected value of an indicator is the probability of the event. So the sample mean estimates $P(X \le 2)$. The exact value is `pexp(2)`.

Question 4

You estimate $\theta = P(X > 3)$ by the sample mean of the indicators $\mathbf{1}(X_i > 3)$. Why is `binom.test` preferred over `t.test` for the confidence interval?

1. `t.test` cannot handle values of 0 and 1.
2. `binom.test` intervals are always narrower.

3. The indicators are Bernoulli, so a proportion interval applies exactly.
4. The indicators are not identically distributed.

Solution

Answer: 3. Each indicator is Bernoulli(θ), so a binomial interval is exact rather than an approximation. `t.test` runs on 0/1 data without complaint, but its interval leans on the central limit theorem. Binomial intervals are also not always narrower.

Question 5

With n function evaluations in k dimensions, grid-based integration has error rate $O(n^{-2/k})$, while Monte Carlo has error rate $O(n^{-1/2})$.

For an integral with 8 dimensions ($k = 8$), which method would be expected to have less error for a very large number of function evaluations (i.e. n is large)?

1. Grid-based
2. Monte Carlo
3. They would be about the same
4. It depends on the function being integrated, not k or n.

Solution

Answer: 2. At $k = 8$ the grid error goes like $n^{-2/8} = n^{-1/4}$, which is slower than Monte Carlo's $n^{-1/2}$, so Monte Carlo wins once n is large. The rates tie at $k = 4$, and the grid is better below that. The integrand affects the constants in both rates, but for large n the exponents decide, so option 4 is wrong.